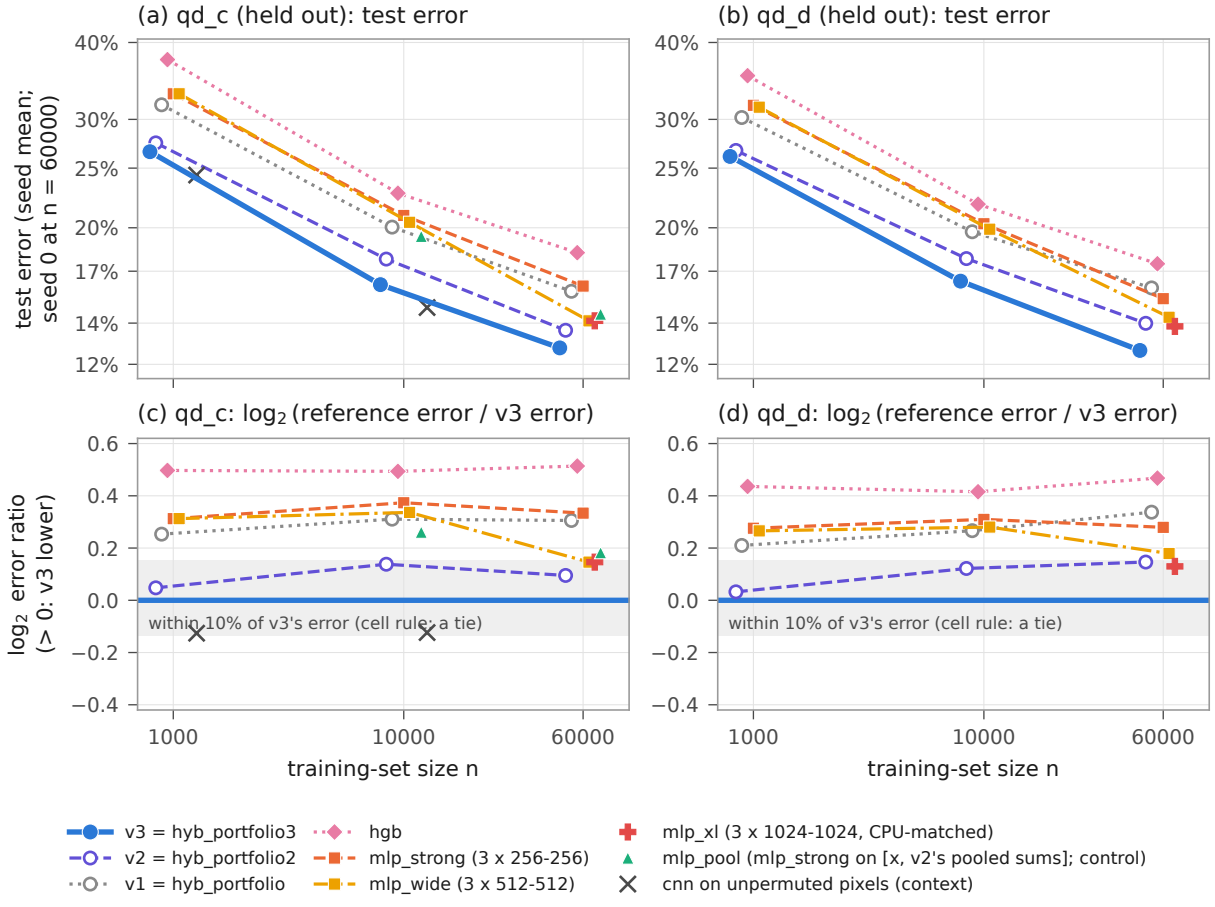
(a) qd_c (held out): test error
(b) qd_d (held out): test error
test error (seed mean; seed 0 at n = 60000)
40%
30%
25%
20%
17%
14%
12%
(c) qd_c: log2 (reference error / v3 error)
(d) qd_d: log2 (reference error / v3 error)
log2 error ratio (> 0: v3 lower)
0.6
0.4
0.2
0.0
−0.2
−0.4
within 10% of v3's error (cell rule: a tie)
1000
10000
60000
training-set size n
v3 = hyb_portfolio3
v2 = hyb_portfolio2
v1 = hyb_portfolio
hgb
mlp_strong (3 x 256-256)
mlp_wide (3 x 512-512)
mlp_xl (3 x 1024-1024, CPU-matched)
mlp_pool (mlp_strong on [x, v2's pooled sums]; control)
cnn on unpermuted pixels (context)